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The Bright Side of the Moon: Transfer Learning and Creativity in Machine Learning Practice

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ABSTRACT

AI systems, such as neural-network-based deep learning (DL) and other machine learning (ML) algorithms, can extract valuable insights from data. A major downside of these algorithms is dependence on the availability of sufficient amounts of relevant and structured data. This is clearly problematic for uses in settings where data are scarce and may hamper the development of innovative, creative ML solutions. Hence, there are tensions between ambitions expressed in previous studies to build “universal” solutions based on available (big) data, and the need to contextualize data for specific uses in distinct domains. However, in this paper, we argue that recent advances in ML reduce these tensions and call for more understanding of how these systems can facilitate human creativity. To contribute to such understanding, we present an illustrative application of transfer learning (TL) to facilitate conceptual leaps by broadening algorithmic affordances. This application, involving the use of data on the shapes of lunar craters to identify archeological remains in Swedish forests, highlights how TL can act as a catalyst for cross-domain idea generation through a practical example. By doing so, we theoretically link research on ML development with creativity research, while also demonstrating this connection in practice.

Introduction

Many modern innovations include neural-network-based deep learning (DL) and other machine learning (ML) algorithms (Jordan & Mitchell, 2015; LeCun et al., 2015; Samtani et al., 2023). In this paper, we present a case that illustrates how data from other domains can be used in creative exploration of new avenues for innovation and problem-solving. In doing so, we establish a theoretical link between research on ML development and creativity research, while also demonstrating this connection in practice.

The ideal inputs for ML algorithms are high-quality datasets that cover wide ranges of variation in particular phenomena, thus allowing them to learn pertinent patterns and generate relevant predictions. However, real-world datasets are often incomplete, inconsistent, or imprecise. Here, two contrasting narratives regarding the role of data have emerged in recent years. Contributors to one narrative (e.g., Ansari et al., 2015; Halevy et al., 2009; McAfee et al., 2012) have argued that increases in “Big Data,” which has become massively more abundant and diverse in recent decades, will automatically drive dramatic improvements in both the scope and performance of ML algorithms and

analytic procedures. Contributors to the other narrative highlight a need to recognize that data are contextually embedded, practice-entangled (Gitelman, 2013; Iliadis & Russo, 2016; Parmiggiani et al., 2022) and must be appropriately contextualized, or re-contextualized, to be useful for training ML models (Domingos, 2012; Jones, 2019; Räsänen & Nyce, 2013). For example, self-driving cars must be capable of identifying multiple species of large animals, which became an issue for Volvo when their systems could not correctly identify kangaroos since their movement patterns differ from those of other animals, such as elks (Piramuthu & Caesar, 2021).

Here we argue that both of the outlined narratives miss the mark. We propose that knowledge can be transferred between domains in innovative ways that enable effective training of ML algorithms for uses in domains and settings where data are insufficiently rich or precise through algorithmic creativity and transfer learning (TL, use of data from one domain in another domain). However, the practical use of data from one domain to solve real-world problems in another domain with contrasting contextual elements is still in its infancy due to its novelty, and there have been calls for more explorations of such approaches (Geiger et al., 2022). In this paper, we investigate how TL

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can alter the affordances (e.g., extend the use of datasets beyond their native domain) of ML algorithms. As an empirical example of this, we aim here to explore the potential utility of TL techniques to broaden understanding of how ML can be used to solve complex problems and foster creativity. Against this backdrop, our research question (RQ) is:

How can transfer learning enable creativity in machine learning practice?

We address this RQ by exploring the use of TL as a source of creativity in ML in a specific case, involving use of data on the shapes of lunar craters to identify archeological remains in Swedish forests. The results highlight TL's potential utility for freeing researchers from the constraints of traditional ML approaches and allowing them to focus on generating creative solutions.

This paper proceeds as follows. [Section 2](#) presents related research on transfer learning, AI and creativity. The methods applied in the exploration of the empirical case are described in [Section 3](#). Results are presented in [Section 4](#), and discussed in [Section 5](#). Finally, conclusions are presented in [Section 6](#).

Related research

To facilitate an understanding and analysis of the relationship between TL and creativity, we position this paper in two literature streams. First, we summarize literature on TL in subsection 2.1. Then, in subsection 2.2, we present research on AI, creativity, and affordances.

From big data to transfer learning

ML development involves experimental practices, and the solutions are often dynamic (Hopf et al., 2023). Meanwhile, increases in computational processing power and access to extensive datasets, commonly referred to as “big data” (Günther et al., 2017; Lycett, 2013), have empowered algorithms to “learn.” This has accelerated AI development since the turn of the millennium (Russell & Norvig, 2022). A paradigm shift was also triggered when Alex Krizhevsky and colleagues won the ImageNet Large Scale Visual Recognition Challenge in 2012 with a neural network-driven ML model called AlexNet (Krizhevsky et al., 2012), which outperformed competitors by a large margin. ImageNet is a large dataset of annotated images that is commonly used as a benchmark for evaluating the performance of algorithms designed for image classification and object recognition. Before ImageNet, practitioners were restricted to algorithms that only required small sets of images due to the paucity of image data on the web and high costs of digital cameras. Soon, large datasets of other

types of digitized phenomena and artifacts were developed, such as music, 3D objects, and text. The latter spurred rapid development of natural language processing (NLP) models, such as BERT and Open AI's GPTseries.

Thus, advances in ML have dramatically increased opportunities for obtaining insights and predictions using algorithms. Following the advent of DL systems, artificial neural networks capable of representing and processing complex features help the detection of patterns in data in many tasks. However, both a strength and weakness of DL systems is their reliance on the availability of data of sufficient quality and structure (Gröger, 2021). This poses major challenges for AI developers and users in many industries and other settings where there are “isolated islands” of data that lack requirements in terms of content, size and/or structure to provide robust foundations for predictive systems (Yang et al., 2019). This is clearly problematic for operations in settings where available data lack sufficient volume and/or suitability. For example, in ecology and forestry much data collection has traditionally relied on extremely time-consuming field inventories that by necessity focus on sampling plots (literally small islands in some cases) covering small proportions of total surveyed areas (Duvemo & Lämås, 2006). Thus, to produce value from ML, organizations must find ways to bridge the differences in scale, scope and limitations between the available isolated islands of data and required datasets.

A potential solution to the problem that is emerging in ML research, especially in DL applications, is TL, which refers to a set of approaches that enable the reuse of previously learned patterns from a source domain to solve new problems in a target domain (see, [Figure 1](#)), thereby helping efforts to overcome challenges related to data scarcity, computational

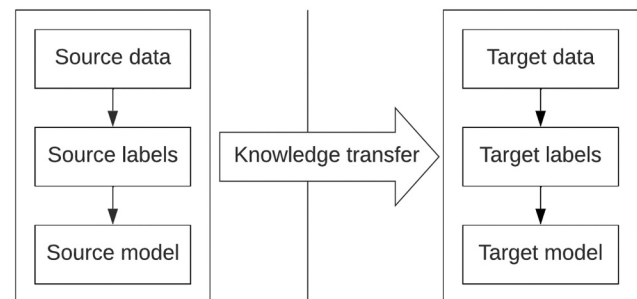


Figure 1. Illustration of transfer learning. In a general ML workflow, a dataset is labeled (e.g., with a text describing what an image or text depicts). This dataset is then used to train a model, to make predictions on previously unseen data. Transfer learning is a collection of methods to transfer “knowledge” from other sources into the ML workflow, to improve the model's performance.

complexity, and the generalization of models (Shao et al., 2014; Tan et al., 2018; Wu et al., 2017).

If done correctly, TL may reduce needs for labor-intensive data-labeling efforts (Pan & Yang, 2009). The use of pre-trained models can improve key elements of diverse tasks, such as image recognition (Shin et al., 2016) and natural language processing (NLP) (Raffel et al., 2020). Examples range from use cases in industrial operations (Li et al., 2022; Shao et al., 2018) and medical imaging (Apostolopoulos & Mpesiana, 2020; Kim et al., 2022) in contexts where relevant data are scarce (Han et al., 2021; Kora et al., 2022) or difficult to access, due to legal protection of healthcare data for instance. Potential downsides of using pre-trained models include possible bias associated with contextual conditions that influenced the training data (Denton et al., 2021), and the behavior of TL models may be unpredictable due to the datasets' size and complexity (Lotte et al., 2018). Moreover, if the source and target domains differ too much, "negative transfer" (Weiss et al., 2016) may also occur, i.e., the models may perform less well in the target domain than in the source domain.

Several types of TL methods have been outlined (Lu et al., 2015; Shao et al., 2014; Taylor & Stone, 2009; Zhuang et al., 2020).

The main approach in DL is *model-based TL*, which involves transfer of knowledge from a model to a target domain (Ravishankar et al., 2016; Chang et al., 2016; see also, Caruana, 1997).

As most DL developmental efforts currently focus on model-based approaches due to their versatility and effectiveness in domain adaptation and handling complex data transformations (Ravishankar et al., 2016; Tan et al., 2018; Yang et al., 2023; Zhuang et al., 2020), we choose to use this method in this paper. As described in the following section, we present a case involving use of data on the size and shape of impact craters on the surface of the moon, to train a deep neural network. The trained model was then transferred to the forestry domain where it was fine-tuned using field data on hunting pits in Swedish forests and data acquired by high-resolution airborne laser scanning (ALS). Our results show that transfer learning of features from the impact craters using the hunting pit data resulted in a higher performing model than training solely on hunting pit data.

Creativity and AI affordances

Previous studies have clearly articulated how creativity plays a key role in innovation processes, for example by approaching problems through analogical thinking and abilities to reconceptualize problem framing and both integrate and reuse features from multiple domains (Malhotra & Majchrzak 2022; Majchrzak et al. 2004; see

also Duverger, 2012; Taylor & Greve, 2006). The use of AI may indirectly lead to an increase in creative work by automating repetitive tasks (Lee et al., 2023). However, emerging research also shows a more direct connection between AI and creativity by exploring interactions between humans and intelligent systems (see, e.g., Fügner et al., 2022; Siemon et al., 2022). Pan and Nishant (2023, p. 6) highlight how human competencies are vital for developing innovative AI solutions to complex problems, as "Human capital with strong creativity skills can develop AI's data-driven recommendations into actionable solutions." Emerging research emphasizes the role of AI in fostering creativity and innovation (Pagani & Champion, 2023). However, until recently, the capabilities of AI systems have been limited to narrow applications (Marrone et al., 2024; see also, Wingström et al., 2022).

As noted by Boden (1998, p. 347), "Creativity is not a special 'faculty,' nor a psychological property confined to a tiny elite. Rather, it is a feature of human intelligence in general. It is grounded in everyday capacities such as the association of ideas, reminding, perception, analogical thinking, searching a structured problem-space, and reflective self-criticism." Boden (2010) recognized three types of creativity: combinatorial, exploratory, and transformational. Combinatorial creativity involves combining familiar ideas in unfamiliar ways, exploratory creativity pertains to exploration within conceptual spaces, and transformational creativity refers to creative alterations of the conceptual space itself. AI can play a crucial but different role in each of these types of creativity. In combinatorial creativity, AI systems can juxtapose ideas from diverse domains, generating novel combinations. In exploratory creativity, AI can traverse a defined conceptual space, potentially uncovering new and interesting ideas within established boundaries. Transformational creativity remains the most challenging for AI as it involves redefining or extending existing conceptual frameworks, tasks that are significantly more complex for AI systems due to their inherent design limitations. In addition, AI-driven creativity lacks the inherent emotional, cultural, and experiential elements of human creativity. While AI can generate outputs that mimic creative works, they are rooted in algorithmic processes and lack the subjective experiences and intentions that characterize human artistry. This distinction raises essential questions about the nature of creativity itself and the role of human experience in creative expression (see e.g., Koivisto & Grassini, 2023).

Creativity has been defined as "the ability to come up with ideas or artefacts that are new, surprising and valuable" (Boden, 2004, p. 1). The term artifact traditionally referred to anything that was man-made, including such diverse things as artworks and technological devices. Artifacts may now also include algorithms, and potentially

anything created either by algorithms or by a combination of humans and AI. In parallel with the development of intelligent systems, scholars have explored how AI can be used to stimulate creativity (Watkins & Barak-Medina, 2024). A particularly interesting research strand here investigates the relation between humans' ability to make "conceptual leaps" by creating surprising and unexpected things from typical knowledge by thinking "outside of the box" (Weisberg, 2010). A conceptual space is a medium where people understand, do and imagine things. However, these spaces also extend to material spaces as they shape designs and interactions between, for example, digital and physical objects (Benyon, 2014). Conceptual spaces may have highly varying restrictiveness, and we discover their limits through explorative activities, but in some cases we may also change, or transform, them (Boden, 2004, pp. 4–6).

The impact of resource scarcity on creativity and innovation has been subject to conflicting findings in previous research. Constraints tend to lead to various effects on how people approach problems and generate solutions (Acar et al., 2019; Caniëls & Rietzschel, 2015; Sonenshein, 2014). In this paper, we explore how TL can stimulate creative solutions in settings where data are a scarce resource. Thus, while one could argue that ML/DL offers rather rigid conceptual spaces due to the demands for structured data, as mentioned in the previous section, TL approaches enable the expansion of such data spaces. Thus, we are interested in how TL changes the affordances (i.e., the perceived possibilities for how ML can be used) (Gibson, 1979) associated with ML, by altering these conceptual spaces. Affordance theory has been successfully used in previous studies to show how technology both constrains and facilitates new opportunities for people to reach their goals (Leonardi, 2011; Seidel et al., 2013). As noted by Veale et al. (2020), computers have been considered valuable for enhancing human innovation for a long time, as evidenced by their widespread adoption in the creative industries. However, although researchers have paid considerable attention to the formation of creative partnerships between humans and machines (Lubart, 2005), until recently these partnerships have been limited by tight restrictions of intelligent systems' capacities. Before the advent of TL methods mentioned in the previous section ML systems could not solve problems without domain-specific training data, and thus were unable to support conceptual leaps due to their limited affordances (Guzdial & Riedl, 2018).

Thus, this study is well aligned with contemporary research devoted to investigating how novel AI systems can be used to augment human thinking by allowing explorations of larger problem-solution spaces compared to previous, more narrow AI systems (Bouschery et al., 2023; Jussupow et al., 2021; Raisch & Krakowski, 2021). While research of the processes involved in joint human-AI

engagement in creative endeavors has increased with the advent of generative AI (Holzner et al., 2025; Rafner et al., 2023; Vinchon et al., 2023), in this paper we argue that there is still much room to explore these issues in relation to ML and TL. Our approach to this exploration is presented in the following section.

Methods

To answer the RQ, *How can transfer learning enable creativity in machine learning practice?*, we investigated the potential of using DL and TL to solve a problem in a domain where data remains scarce: the detection of cultural remains of "hunting pits" in Swedish forests. In this section, we first describe the case and domain we explore (Subsection 3.1), then two approaches to identify the pits using DL techniques (with and without TL). This study is situated in the intersection between data science and social sciences, as our approach heavily relied on experiments in this domain, but our results are also of a processual nature, as we describe in detail our considerations, and their relationship to the notion of creativity.

The case: detecting hunting pits using deep learning

Growing demands for stronger legislation and less aggressive forestry methods, fueled by increasing pressure on the world's forests and increasing knowledge of past human land use (including by indigenous peoples) have fostered international obligations to protect cultural remains (Agnoletti & Santoro, 2018). Consequently, in accordance with Article 1 of the United Nations' Convention Concerning the Protection of the World Cultural and Natural Heritage (United Nations, 2023), Sweden is obliged to protect ancient monuments and cultural remains in forests, particularly those considered unique to Sweden or northern Scandinavia, such as hunting pits. Hunting pits are deep pits that were dug in the ground to trap animals such as elk, reindeer, wolves and bears. They were used from the Stone Age until the 19th century. The pits were camouflaged using branches and leaves, and animals that fell into one were either trapped or killed by sharpened sticks placed in the bottom of the pit.

Investigations have repeatedly found that forest owners often fail to protect these cultural remains, and too many are damaged by forestry operations. The main reasons for this are a lack of functional information, deficiencies in machine operators' training, and logging during winter, when the remains are covered by snow. Moreover, current maps of hunting pits are incomplete and have poor spatial accuracy. To address this issue, we investigated the possibility of automatically mapping the pits using national Swedish airborne laser scanning data

and DL. Trier et al. (2021) have previously used a Region-based Convolutional Neural Network and dense ALS data to map charcoal kilns, hunting pits, and grave mounds in parts of southern Norway. Segmentation approaches can also be used to map archeological features (Ibrahim et al., 2019). In this study we used a segmentation approach based on a UNet architecture (Ronneberger et al., 2015) to map small-scale archeological features from national ALS data (Figure 2). We focused on hunting pits since they have proven difficult to map using scarce ALS point clouds and are unique to northern Scandinavia. For this, we introduce a dataset with 2519 manually mapped hunting pits spread across 1275 km² in a landscape mainly dominated by forest.

A well-known constraint in archeological prospection based on remote-sensing (and many other efforts) is the lack of ground-truth samples (training data). We evaluated two strategies to address this limitation: data augmentation, and TL. Coordinates of known hunting pits were downloaded from the Swedish National Heritage Board (Figure 3). The coordinates were manually adjusted based on visual observations in the ALS data. The area surrounding known hunting pits was inspected and additional hunting pits were manually digitized if discovered. In total, 2519 hunting pits were digitized in this way. These polygons were converted into segmentation masks where pixels inside the polygons were given a value of 1 and pixels outside the polygons were given a value of 0.

Analytical approach

We made two attempts to detect hunting pits, which are described in more detail in the results section. Links to repositories containing the code and dataset can be found in Lidberg et al. (2024). During the first attempt we trained a DL model using TensorFlow 2.6 on ALS data processed using digital

elevation models (DEMs, see Figure 4), in which positive profile curvature values indicate acceleration of flow across the surface, while negative values indicate deceleration of flow (Shary et al., 2002). DEMs with 0.5 m resolution were created from the ALS point clouds using a tin gridding approach implemented in Whitebox tools 2.2.0 (Lindsay, 2016). The images were reduced in size, and augmented through rotation and flipping to increase the number of images in the training data.

During the second attempt, we employed a TL approach using data on lunar impact craters from NASA's Moon craters database (Robbins, 2019) and a DEM generated from data obtained by The Lunar Orbiter Laser Altimeter on the Lunar Reconnaissance Orbiter (LRO) spacecraft (Mazarico et al., 2012). These datasets were used to pre-train a model with the crater data before fine-tuning it on the hunting pit dataset (with a random split of the data, using data on 80% of the craters for training and 20% as a test set for evaluation purposes). Then we assessed the transferability between the source and target datasets in terms of the "performance of the second learner with learning from the first learner minus the performance of the second learner without learning from the first learner" (Weiss et al., 2016, p. 30) using F1 scores. The F1 score is a balanced performance measure that accounts for both false positives (features incorrectly identified as hunting pits) and false negatives (failures to correctly identify actual hunting pits).

Results

In this section we describe our considerations and results regarding the detection of hunting pits both without (Subsection 4.1) and with the lunar data (Subsection 4.2). Then, we present evaluations of both attempts (Subsection 4.3).



Figure 2. A field-inventoried hunting pit (left) and an image of the same pit derived from the ALS data (right), in which the sides of each square represent 50 cm in the real world.

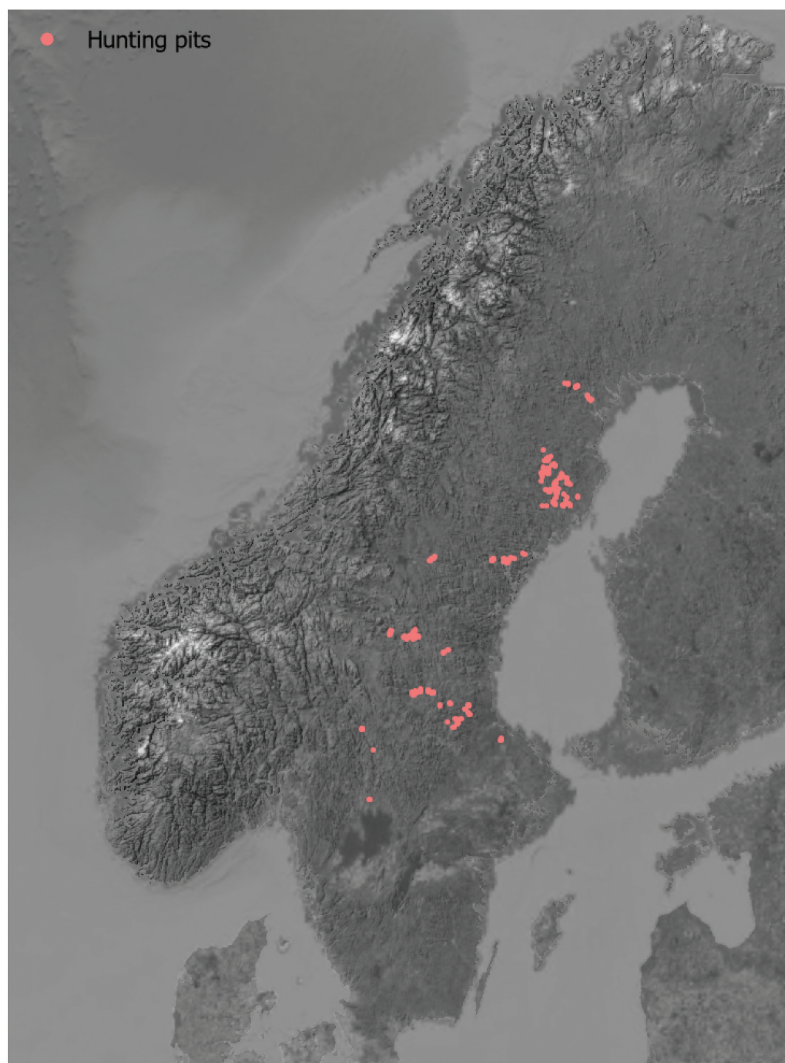


Figure 3. Locations of hunting pits used in this study. In total, 2519 hunting pits were manually mapped in Sweden.

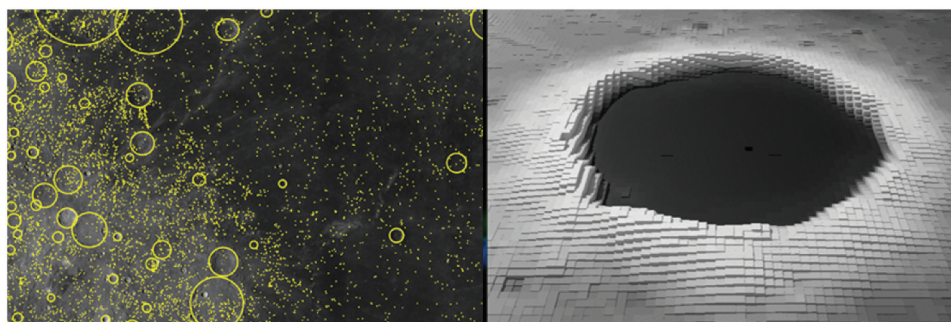


Figure 4. Examples of digitized impact craters in the lunar DEM (left image) and an enlarged image of one of the craters (right image, where each square corresponds to 118 x 118 m).

Attempt 1: automatic detection of hunting pits using airborne laser scanning data

Ideation

The opportunities to map archeological remains using data obtained by remote-sensing technologies are rapidly advancing, with increases in both the amounts and quality of available data. In this study we attempted to map hunting pits using ALS data to extend the limits of the available data in Sweden. Hunting pits are small-scale features that are challenging to detect from aerial imagery and the current ALS data only contained 1–2 laser pulses per square meter. However, previous work on mapping small-scale features such as charcoal kilns and ditches in the Swedish landscape inspired us to test the feasibility of using similar ALS-based methods to detect and map hunting pits.

Data collection

A compact laser-based system (Leica ALS80-HP-8236) was used to collect the ALS data from an aircraft flying at 2888–3000 m. The ALS point clouds were divided into 204 tiles, each with a size of 2.5×2.5 km, covering an area of 1 275 km² in total.

Data processing

Profile curvature values from the DEM were calculated to help the DL model to learn the appearance of hunting pits in the landscape. Profile curvature is a common measure to detect variances (e.g., slopes) in geographical analysis. The 204 tiles were split into smaller image chips with 250×250 pixels per chip. Image chips without hunting pits were removed to combat the highly imbalanced class distribution.

Data augmentation

We assumed that the hunting pits and terrain immediately surrounding them did not have any preferred orientation, which allowed for image augmentation by rotating and flipping the image chips during training. This is a common data augmentation technique used to increase the number of images in a dataset.

Model training

TensorFlow 2.6 (an open source ML framework supplied by Google) was used to build a UNet, which is a deep neural network, to transform the profile curvature images into images highlighting the detected hunting pits.

Attempt 2: using lunar data to identify hunting pits

Ideation

The original model did not perform as well as we had hoped, prompting us to think about ways to

improve the model and data. Experimenting with different models did not improve the results, and digitizing more training data would have been very time-consuming and costly given the scarce sources. We then started to examine how researchers in other disciplines had tackled similar problems in cases with scarce data and identified TL as a potentially effective strategy to expand the affordances offered by DL. The challenge was to find an available large-scale digitized high-resolution ALS dataset representing features with sufficiently similar shapes to hunting pits. After combing through terrestrial databases, we found NASA's database of lunar impact craters. These are much larger than hunting pits (more than 1 km in diameter), but the Lunar DEM pixels are correspondingly larger than those obtained with our ALS data. Thus, the lunar crater dataset warranted consideration for use in our TL strategy, as it enabled a conceptual leap between two unrelated domains.

Data sourcing

The basic TL strategy was to regard impact craters as pits, with sufficient similarity to hunting pits on Earth (despite the difference in scale). The crater images were digitized by NASA and obtained from the Moon Crater Database v1, which includes data on approximately 1.3 million lunar impact craters and is almost complete for all craters larger than about 1–2 km in diameter.

Model pre-training

The DL model was pre-trained with the lunar data before being trained on the hunting pit dataset.

Evaluation

80% of the image chips were used to train the models and 20% to test them, which is a common division in ML practice. The test dataset was used to calculate numbers of true positive, false positive and false negative predicted pixels based on common measures such as recall, precision and F1 score, which indicate the performance of a model on a scale between 0–1. The F1 score for the first model (from 4.1 above) was 0.56 and use of the lunar crater data (from 4.2 above) improved this to 0.62, with an increase in recall value from 0.45 to 0.52. [Figure 5](#) shows the differences in F1 scores between the attempts, and [Figure 6](#) shows an example of differences in predictions between them.

In sum, our research process highlights how TL facilitates conceptual leaps, as illustrated in [Figure 7](#), and further outlined in the following section.

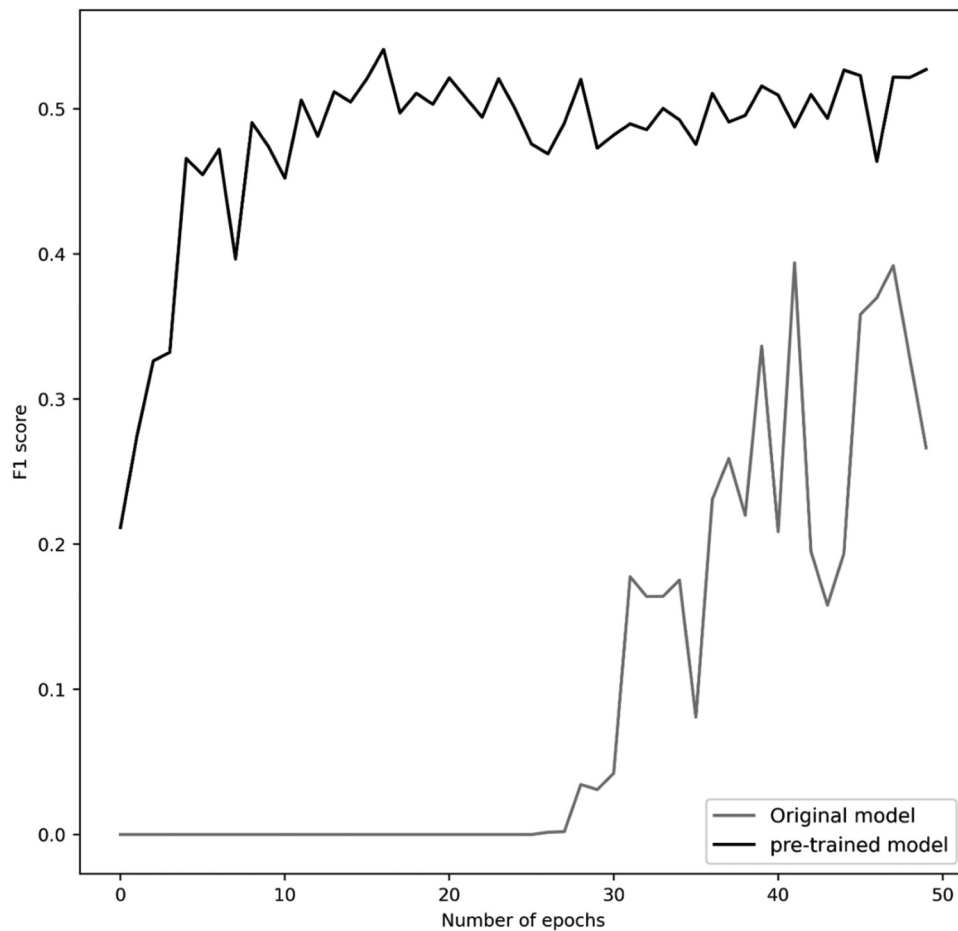


Figure 5. Curves visualizing the learning process over time obtained from the evaluation of the models on the test dataset. Each time a dataset is processed by a ML algorithm is called an epoch (x-axis). If a model manages to reach a good F1 score (y-axis) early, it is usually a good sign. The original model (without lunar craters) struggled to find appropriate weights during the first epochs and was less accurate than the model trained on lunar craters, which immediately found approximate weights and quickly started to improve them.

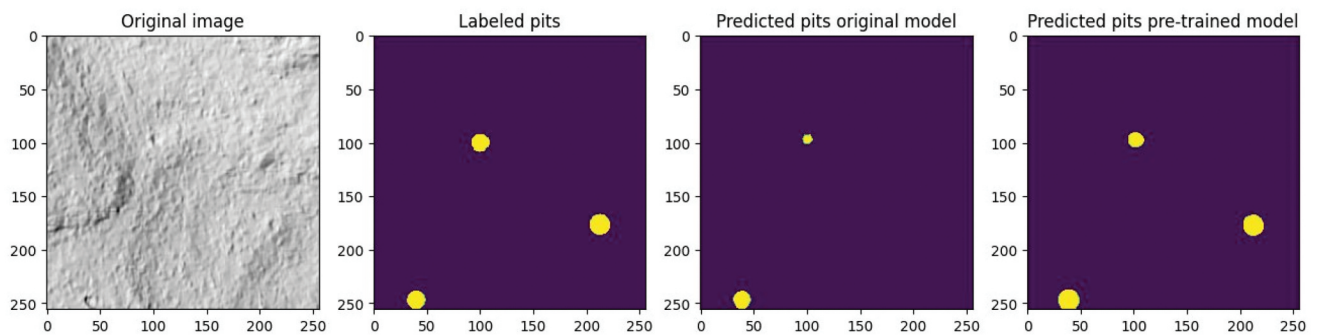


Figure 6. An image from the test dataset showing the labeled and detected hunting pits (yellow dots in the image). The left image is the training data, the middle the prediction of the original model (without lunar craters) and finally on the right is the prediction from the model that was pre-trained on the lunar crater data.

Discussion

To extend the traditional boundaries of AI by applying algorithmic creativity the systems must be capable of not only learning and applying knowledge but also

making conceptual leaps (Weisberg, 2010) across disparate fields. To contribute to such efforts and understanding of the processes involved we have addressed the RQ *How can transfer learning enable creativity in*

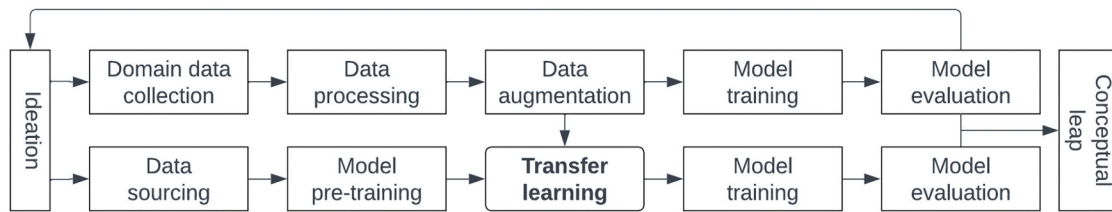


Figure 7. Illustration of the research process. The top row represents the first attempt, without lunar crater data where we looped back to an ideation stage due to poor model performance. The bottom row illustrates the inclusion of TL and lunar data, which generated an improved model.

machine learning practice?, focusing particularly on application of TL to identify hunting pits using data on lunar craters.

Algorithmic creativity in transfer learning

The use of crater data to identify hunting pits represents a conceptual leap, showcasing the ML model’s capability to re-use (Malhotra & Majchrzak, 2022) knowledge gained from one domain (lunar landscapes) to another (terrestrial archeology). This leap was facilitated by TL, which allowed the model to apply its understanding of crater-like formations to effectively identify hunting pits, thereby highlighting several important aspects of creativity in human-AI interaction. TL essentially widened the affordances associated with ML algorithms, enabling not only extended exploration within the focal conceptual space (i.e., boundaries of the domain), but also combination of different concepts (Boden, 2010) to generate a new and valuable artifact in the form of an improved ML model (Boden, 2004), by enabling conceptual leaps (Weisberg, 2010), as illustrated in Figure 8, and further outlined below.

Here, our study shows how ML algorithms’ rigidity can be reduced via the increased “algorithmic” affordances (cf., Gibson, 1979) provided by TL, by enabling new perceptions of how ML can be used. These new perceptions (i.e., using more ambiguous forms of data to approach a problem) effectively extend humans’ conceptual spaces in relation to ML and data. Taken together, by combining theories of affordances (Gibson, 1979) and conceptual spaces (Boden, 2010), we show how TL can be used to make conceptual leaps (Weisberg, 2010), which has important implications for creativity research.

Theoretical implications

Our findings contribute to the broader discourse on AI and creativity. We have demonstrated how AI can support human explorations of new concepts, and engage in combinatorial creativity by enabling the merger of

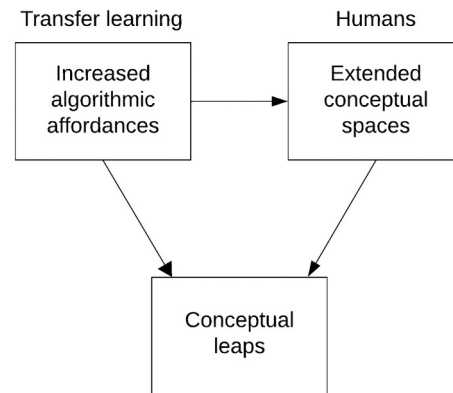


Figure 8. An illustration of how transfer learning facilitates conceptual leaps. Taken together, the increased algorithmic affordances offered by TL, enable humans to extend the conceptual spaces in relation to machine learning, which leads to conceptual leaps (manifested in this study by using data from a different domain (moon craters) to identify archeological remains (hunting pits)).

ideas from diverse domains (Boden, 2010; see also Wingström et al., 2022). Here, we cross-pollinate theories of affordances (Gibson, 1979) with creativity research on conceptual spaces (Boden, 2004, 2010) and conceptual leaps (Weisberg, 2010). By doing so, this study provides two major contributions:

First, our study signposts the potential utility of TL in the creative application of ML. By transcending traditional domain boundaries in the use of lunar crater data to address terrestrial challenges like hunting pit detection, we have shown how AI can facilitate human creativity by extending algorithmic affordances and enabling novel combinations of concepts. This fusion of AI’s computational power with human ideation enables the generation of innovative solutions. We demonstrate how AI can become a partner in creative endeavors, by aiding combination of ideas, exploration, expansion and transformation of conceptual spaces (Boden, 2004; Bouschery et al., 2023; Guzdial & Riedl, 2018), which resonates well with contemporary scholarly work on “hybrid” intelligence (see Davis et al., 2025).

Second, we bridge the gap between narratives regarding the roles of big data (Ansari et al., 2015; Halevy et al., 2009; McAfee et al., 2012) and situated data (Domingos, 2012; Jones, 2019; Räsänen & Nyce, 2013). Our research demonstrates that ML's reliance on data can be, at least partially, satisfied by the use of TL. Here, we show that ML can be effectively trained on "universal" datasets like the lunar crater dataset, then adeptly applied to context-specific tasks such as identifying hunting pits. This harmonization of big and situated data challenges the traditional narrative that ML/DL requires domain-specific training to be effective. This not only broadens the applicability of ML, but also greatly extends its role as a creative tool in diverse contexts. In doing so, our contribution distinguishes itself from contemporary research on AI and creativity, which often focuses on generative AI and large language models (Holzner et al., 2025; Rafner et al., 2023; Vinchon et al., 2023). Our study shows that there remains considerable room to explore creative uses of ML, particularly in scenarios where data are scarce.

Implications for creativity research

Our approach is a direct response to previous calls to investigate practical uses of novel TL techniques in new domains (Geiger et al., 2022). The practical implications of this research extend beyond lunar craters and hunting pits. The presented methodology can be applied in various fields, such as environmental monitoring, where AI could identify changes in landscapes indicative of ecological shifts, or urban planning for detecting infrastructural anomalies. The versatility of this approach highlights its potential utility in many sectors. Here, our study aligns with emerging views of "AI as a craft" (Hopf et al., 2023), showing how the scope of such development can be extended by tools and methods that allow for a wider range of affordances than their predecessors. By doing so, our study offers contributions to creativity research. First, while advances in generative AI technologies have spurred an increased interest in how these systems can be used to support human creativity, we argue that there is still much unexplored potential in classical ML. Here, we exemplify how ML practice due to its inherent limitations (being dependent on structured data of good quality) encourages researchers to find creative ways to overcome such limitations. Second, we show how the combination of ML and TL constitutes a novel area for creativity researchers interested in investigating how creativity is manifested in settings with scarce resources (in our case, data scarcity) (Acar et al., 2019; Caniëls & Rietzschel, 2015; Sonenshein, 2014).

Conclusion

This paper set out to answer the RQ: *How can transfer learning enable creativity in machine learning practice?*

By utilizing lunar crater data to aid the identification of archeological hunting pits, we demonstrate how TL supports making conceptual leaps between previously unrelated domains. The results demonstrate TL's potential to overcome challenges posed by incomplete, inconsistent, or imprecise datasets. As shown by the process model outlined in this paper, researchers and practitioners can harness TL's potential to create more creative and innovative ML solutions that can be applied to solve problems and add value in diverse domains. Thus, the study provides a significant step toward understanding AI's potential for bridging disparate domains through algorithmic creativity and TL. By demonstrating AI systems' ability to repurpose knowledge in novel contexts, it extends conventional boundaries in the application of AI. As we continue to explore this exciting frontier, it is important to foster a collaborative environment where AI and human expertise coalesce, driving innovation and exploring its limits.

Limitations and further research

This study explored the relationship between ML, TL, and creativity. Two limitations apply, related to the experimental setup, and the implications of the study, as outlined below.

First, the application of ML and airborne laser scanning data in archeology presents various challenges and opportunities. A key issue is the risk of misidentification by the algorithms, leading to natural features being mistaken for cultural remains or vice versa, misinterpretations, and potential losses of significant archeological sites. While our approach was effective, it is important to note that these methodologies may not be universally applicable across different domains, and appropriate adaptations may be needed for specific applications and settings, including checks and guard rails for potential transfer of biases across domains (see, Salman et al., 2022). Nevertheless, the possible benefits are clearly illustrated by our model's ability to provide information that is useful in forestry planning and can be directly incorporated into maps already used in logging and soil scarification operations. Moreover, using TL with lunar crater data improved the performance of our model(s), which is encouraging for similar applications where training data are scarce and/or expensive to digitize. For example, the images of impact craters used in this study could be inverted to create

a landscape of mounds, which could then be potentially used to pre-train models intended to map burial mounds or charcoal kilns.

Second, while this research opens several avenues for future studies, including explorations of the application of TL and creativity in other domains, it is limited to one specific use case. Thus, we encourage future research on creativity to further explore how these technologies can stimulate creative behavior, especially in contexts where data availability is limited. We provide one such example in our study, and we see great opportunities to further study these types of initiatives, for example in organizational settings.

We especially encourage scholars to investigate such endeavors in more longitudinal studies, to more carefully outline the dynamics between AI and human creativity.

Disclosure statement

No potential conflict of interest was reported by the author(s).

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